

VARDAAN BAJAJ

AI Engineer | Applied Deep Learning & Full-Stack Systems

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LinkedIn

GitHub

Portfolio

PROFESSIONAL SUMMARY

AI and Systems Engineer experienced in building real-time digital signal processing (DSP) pipelines in C, low-latency deep learning inference microservices, and browser-native query engines. Published IEEE author in aerial computer vision and web architecture with core technical depth across PyTorch, TypeScript, FastAPI, C11 atomics, and distributed data modeling.

TECHNICAL SKILLS

Languages: Python, C/C++11, TypeScript, Modern JavaScript, SQL (PostgreSQL, MySQL), Dart, Bash

AI, Machine Learning & CV: PyTorch, Torchvision, Scikit-learn, XGBoost, YOLOv12, SAHI, ESRGAN, DeepSORT, OpenCV

Backend, Systems & Protocols: FastAPI, Flask, Linux (CLI/Hyprland), Docker, C11 Atomics, Serial Protocols (RS-422, SPI), Deterministic Bus Protocols, WebSockets

Frontend Architecture: React 18, Vite, Zustand, Dexie.js (IndexedDB), TanStack Virtual, Power BI, Tableau

EXPERIENCE

Defence Research and Development Organisation (DRDO)

Hyderabad, India (Hybrid)

Research and Development Intern

Jan 2026 – Jun 2026

- Engineered a real-time digital signal processing (DSP) simulation framework in **C** to process dynamic time-series telemetry and high-frequency data communication links.
- Implemented an in-place **Radix-2 FFT** and spectral peak-detection routine to analyze signal parameters from simulated chirp waveforms under AWGN.
- Modeled **RS-422** ring buffers and a deterministic command-response serial bus **Remote Terminal** state machine with **C11 atomics** for lock-free data integrity.

AgryBin

Remote

AI Intern

May 2025 – Aug 2025

- Engineered deep learning segmentation/classification pipelines in **PyTorch**, processing **50K+ geospatial satellite tiles**.
- Deployed scalable inference microservices via **FastAPI + Flask**, sustaining **<120ms latency** for crop analytics.
- Achieved a **17% accuracy gain** across multi-spectral datasets using transfer learning and dynamic data augmentations.

Mahyco

Remote

Data Science Intern

Aug 2024 – Dec 2024

- Developed a **YOLO-based crop yield estimation pipeline** processing **20K+ field images** with custom real-time preprocessing.
- Constructed interactive spatial yield dashboards and automated backend report pipelines, improving decision efficiency by **23%**.
- Streamlined agronomic data workflows by integrating multi-source operational metrics into structured Python and cloud storage services.

FEATURED PROJECTS

DataVista: Browser-Native BI & Query Engine | React 18, TypeScript, Zustand, Dexie.js, TanStack Virtual

- Built an offline-first BI platform executing sub-millisecond data slicing on **100MB+** client datasets without backend dependencies.
- Constructed a recursive-descent **AST formula parser** for Excel calculations and window functions (ROW_NUMBER, RANK).
- Designed a multidimensional aggregation engine integrated with **TanStack Virtual** for lag-free rendering of dense tabular views.

NeuroInsight-AI: Vocal Biomarker Parkinson's Prediction | Python, XGBoost, Scikit-learn, Pandas

- Researched non-invasive dysphonia detection across 198 vocal profiles using nonlinear dynamics and acoustic perturbation features.
- Formulated the composite **Feature Performance Index (FPI)** combining fractal scaling (DFA), complexity (D2), and pitch variance.
- Trained an **XGBoost** ensemble classifier achieving **96.52% accuracy** outperforming SVM baselines for early-stage screening.

Employee Attrition Analytics Dashboard | SQL (MySQL), Power BI, Tableau, Python

- Normalized and joined **50,000+** HR records across relational schemas to isolate turnover drivers (overtime, compensation stagnation).
- Authored conditional aggregation SQL queries segmenting income cohorts and pinpointing high-attrition flight-risk groups.
- Delivered dual interactive business dashboards in **Tableau** and **Power BI** alongside a predictive candidate churn classifier.

PUBLICATIONS

V-Surveillance: A Hybrid Deep DL Framework for Real-Time Aerial Surveillance Using Drone Imagery

IEEE International Conference on Information and Communication Technology (CICT)

Feb 2026

Published

An Enhanced Object-Oriented Programming-Based Web Page Linker

IEEE Conference on Interdisciplinary Approaches in Tech. & Management for Social Innovation (IATMSI)

Apr 2024

Published

EDUCATION

International Institute of Information Technology, Naya Raipur

Bachelor of Technology (B.Tech) in Data Science & Artificial Intelligence

80.7% (CGPA: 7.57 / 10.0)

Nov 2022 – July 2026